CheckIn

Deployment Report — Docker, CI/CD and Release v2.0.0

Student Attendance System · IDE201 Capstone · Group 2 · College of Science and Technology

Prepared by	Ugyen Norbu (GitHub: UgayNobu) — repository owner, responsible for deployment
Docker / CI work by	Damchey Lhendup (branch feature/ci-cd-docker), reviewed, fixed and merged by Ugyen
Date	Wednesday, 28 October 2026
Repository	github.com/UgayNobu/Checkin (branches: dev and main)
Pull requests	#52 photo fix · #53 dev → main · #54 Docker + CI/CD · #55 Render deploy · #56 release · tag v2.0.0

1. Summary

CheckIn is now live on the internet. The frontend runs on Vercel, the backend API on Render, and the data stays in our Supabase database. Students and tutors can open the app on any phone or laptop, and it can be installed on a phone like a normal app.

Before deploying, I reviewed Damchey's Docker and CI/CD branch. The design was good, but the local database setup would have failed on our current schema and the branch was 11 days behind dev. I fixed those problems, added his unit tests to the pipeline, and connected GitHub Actions to Render so the backend only goes live after every check passes. Both dev and main are now protected: nothing can be merged unless all three checks are green.

5	41	3	6 / 6	v2.0.0
pull requests merged	unit tests in CI	required checks	pipeline runs green	released

What	Address
The app	https://checkin-cst-group2.vercel.app
Backend API	https://checkin-backend-gzbg.onrender.com (health check: /health)
Release	github.com/UgayNobu/Checkin/releases/tag/v2.0.0

2. How the live system fits together

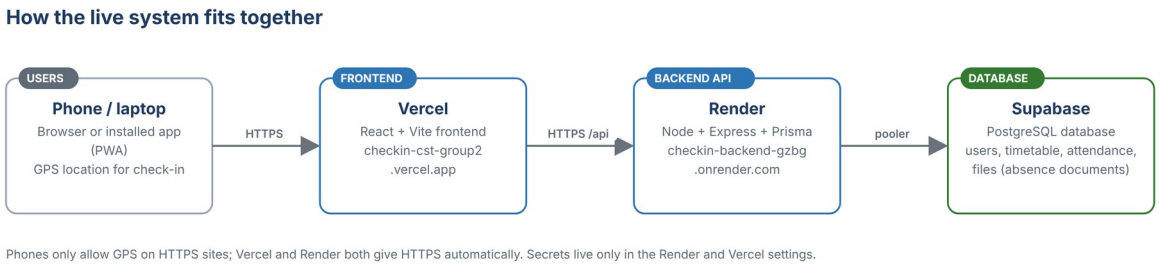


Figure 1 — Phone → Vercel → Render → Supabase, all over HTTPS

Part	Platform	Why
Frontend (React + Vite, PWA)	Vercel	Named in our UAT plan; never sleeps, so the page opens instantly; free HTTPS; a preview link for every pull request
Backend API (Node, Express, Prisma)	Render (free, Singapore)	Runs a normal Node server close to Bhutan; set up from render.yaml in the repo
Database and files	Supabase (PostgreSQL)	Our existing database — no data was moved or changed for the deployment

Phones only allow GPS on HTTPS sites, so check-in works on the live links but not on a laptop's local network address.

3. Review of the Docker / CI branch

Damchey pushed his work as feature/ci-cd-docker. I checked every file, tested the database steps on a fresh PostgreSQL database, ran the frontend build and the tests, and merged the latest dev into it.

3.1 What the branch adds

File	Purpose
backend/Dockerfile	Multi-stage Node 22 image; runs as a non-root user; built-in /health check
frontend/Dockerfile + nginx.conf	Builds the React app and serves it with nginx; forwards /api to the backend
docker-compose.yml	Whole app on one machine: Postgres, database setup, backend, frontend on localhost:8080
docker-compose.prod.yml	Production version that uses the published images and Supabase
.github/workflows/ci-cd.yml	The CI/CD pipeline (section 5)
render.yaml, frontend/vercel.json	Hosting settings for Render and Vercel
PWA files (manifest, icons, sw.js)	Makes the app installable on Android and iPhone; never caches /api data
scripts/smoke-test.sh	Starts the stack and checks that login really works
backend/tests, frontend/src/__tests__	41 unit tests: geofence, distance, check-in codes, enrolment keys, passwords, rooms, form validation

3.2 Problems found and fixed

#	Problem	Fix
1	Fresh database would not build: the schema uses current_academic_session() and seq_class_changes, which the bootstrap did not create (reproduced the error in Postgres).	Added both to db-bootstrap.sql
2	Seed data would fail: tutor assignments and enrolments need a current semester.	New db-after-push.sql adds AS2026 as current, plus 4 unique indexes Prisma cannot create
3	db-legacy.sql re-added the academic_year column we removed from Supabase.	File removed
4	Geofence set to 10 m; accuracy, network grace and email settings missing.	40 m geofence, 150 m accuracy, 120 s grace and SMTP settings in compose and Render
5	Smoke test logged in as U001, which is not in our data.	Uses CST-ICT01
6	3 building tests failed: roomKey was not exported.	Exported; all 41 tests pass
7	Branch 11 days behind dev, with conflicts in App.jsx, base.css, api.js and README.	Merged dev; kept the VLE design; automatic logout on expired login now goes to the landing page
8	On Windows, Docker could not read files inside OneDrive.	Documented: keep the project outside OneDrive

4. Running everything with Docker

Anyone with Docker Desktop can run the whole system without installing Node or PostgreSQL:

```
cp .env.example .env        # then set JWT_SECRET to a long random text
docker compose up --build
# open http://localhost:8080   ·   log in as CST-ICT01 / Test@1234
```

This uses its own local database with test data, never Supabase. The setup script refuses to run against a Supabase address.

5. CI/CD pipeline (GitHub Actions)

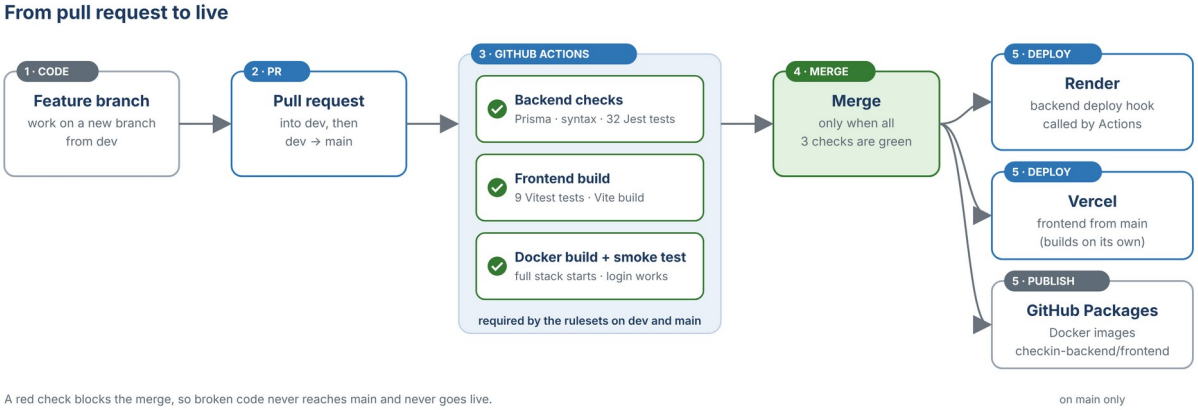


Figure 2 — every change is checked before it can be merged, and only main goes live

Job	When	What it does
Backend checks	Every PR and push	Install, validate the Prisma schema, syntax-check every file, 32 Jest unit tests, security audit
Frontend build	Every PR and push	9 Vitest unit tests, then a production build
Docker build + smoke test	After the two above	Builds all images, starts the full stack and logs in through the API
Deploy backend to Render	Push to main only	Calls the Render deploy hook (a GitHub secret) — only when all checks passed
Publish images	Push to dev / main	Saves the Docker images to GitHub Packages (checkin-backend, checkin-frontend)

Render's own auto-deploy is turned off, so GitHub Actions is the only thing that can update the backend. Vercel builds the frontend from main; if its build fails, the old version stays live.

6. Hosting set-up

6.1 Backend on Render

- **Created from render.yaml** (Blueprint, branch main, free plan, Singapore). Service: checkin-backend.
- **Settings entered in Render, not in the code:** DATABASE_URL and DIRECT_URL (Supabase pooler), FRONTEND_URL, SMTP host, user and app password. JWT_SECRET was generated by Render.
- **Deploy hook** saved in GitHub as the secret RENDER_DEPLOY_HOOK.
- **Checked:** /health answered "ok", and /api/programs returned our programmes from Supabase.

6.2 Frontend on Vercel

- **Project checkin-cst-group2,** root folder frontend, Vite preset, one setting: VITE_API = the Render address + /api.
- **Production branch: main.** Other branches and pull requests get their own preview links.
- **Checked:** logged in as a tutor on the live site; the dashboard loaded the real modules and students. Chrome offered "Install", so the PWA works.

7. Branch protection

Two rulesets protect the team's branches. A pull request is required, force pushes are blocked, and these three checks must pass before merging:

Ruleset	Branch	Required checks
main-protection	main	Backend checks · Frontend build · Docker build + smoke test
dev-protection	dev	Backend checks · Frontend build · Docker build + smoke test

"Require branches to be up to date" is left off on purpose: main has merge commits that dev does not, so it would block every dev → main release.

8. Release v2.0.0

PR	Into	What it did
#52	dev	Sanskar's profile photo fix (with my port fix)
#53	main	Brought main up to date with dev before the Docker work
#54	dev	Docker, CI/CD, PWA and unit tests (Damchey), with the fixes in section 3.2
#55	dev	Backend deploys to Render from GitHub Actions after all checks pass
#56	main	Release; tagged v2.0.0 and published as a GitHub Release (latest)

9. Security

- No passwords or keys are in the repository (it is public). They live only in Render, Vercel and GitHub secrets, and in each person's own .env file.
- The database is never changed by the pipeline: the Docker setup only builds a separate local test database.

> **To do:** the Gmail app password appeared in a screenshot during set-up. Create a new one in the Google account, then update SMTP_PASS in Render and in backend/.env.

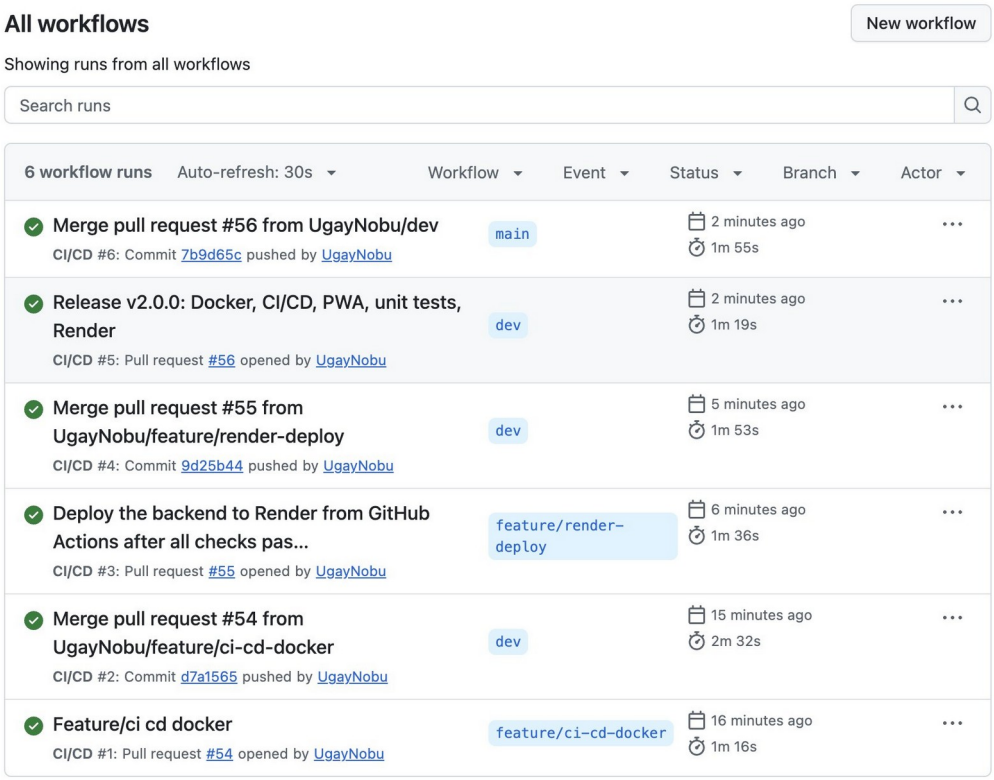
10. For the team

Task	How
Get the latest code	git checkout dev then git pull origin dev
Start new work	Make a branch from dev, push it, open a pull request into dev
Merging	Wait until all 3 checks are green; a red check shows the reason under Details
Release	Open a pull request from dev into main; after merging, Render and Vercel update by themselves
Before a demo or class	Open the /health link a minute early — the free Render plan sleeps after 15 minutes and takes about 50 seconds to wake
Windows + Docker	Keep the project outside OneDrive (for example C:\dev\Checkin)

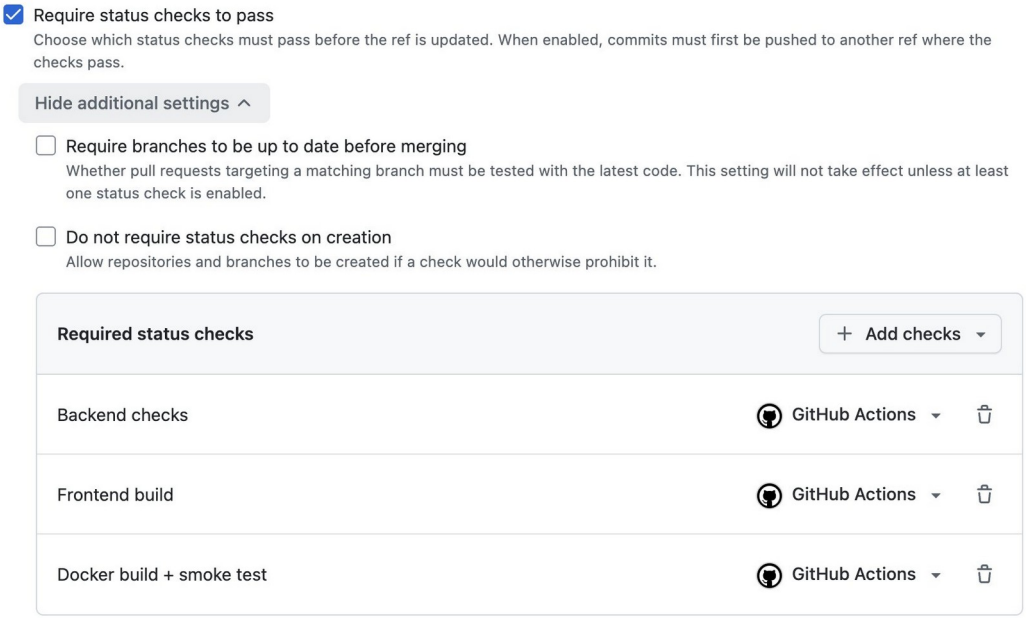
11. Next steps

- Test a real check-in with two phones on the live site: tutor starts attendance, student checks in with location on.
- Replace the Gmail app password (section 9).
- Turn on two-factor login for the Vercel and Render accounts.
- Clean up the old feature branches on GitHub (47 branches).

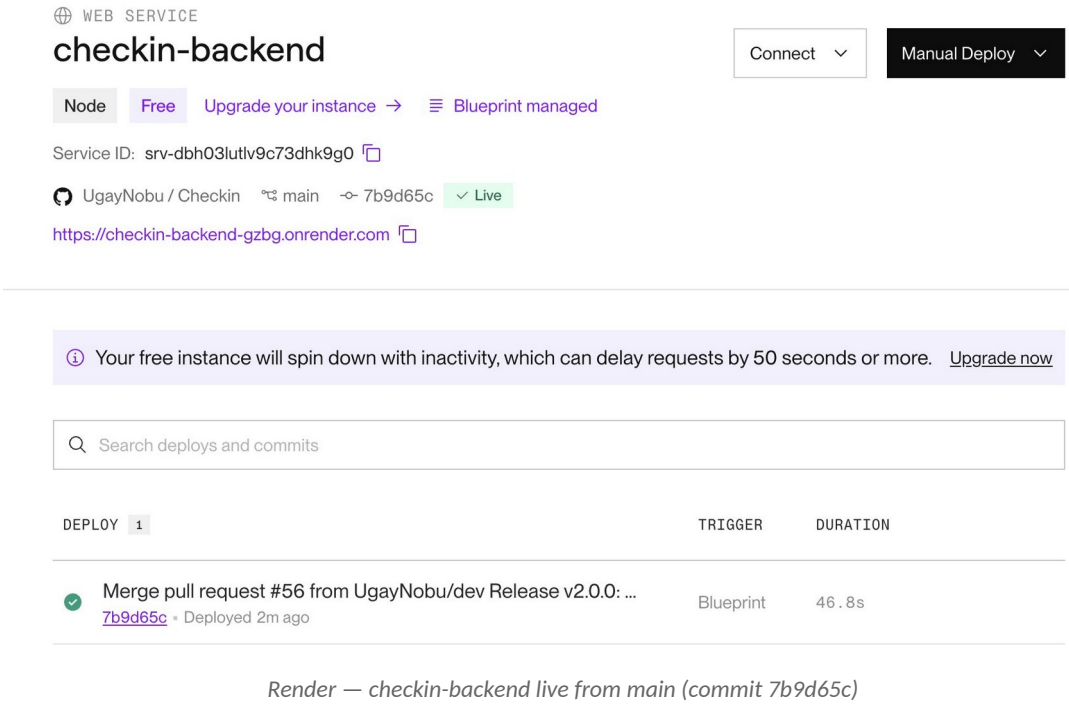
Appendix: Screenshots



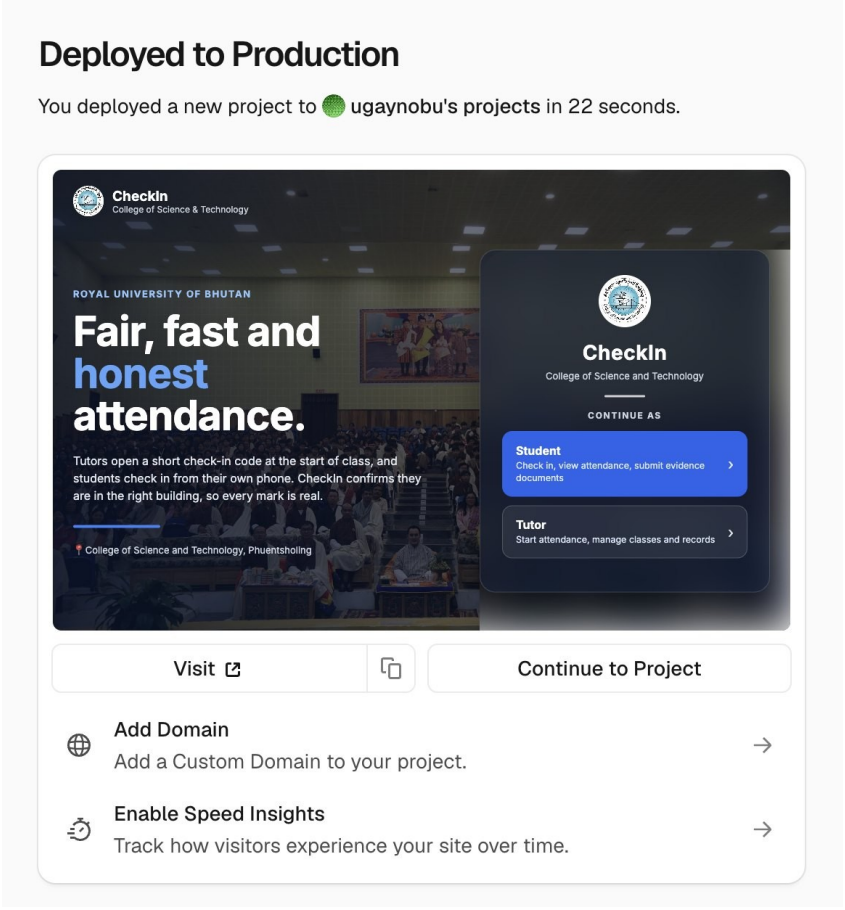
GitHub Actions — all 6 pipeline runs passed, including the release on main



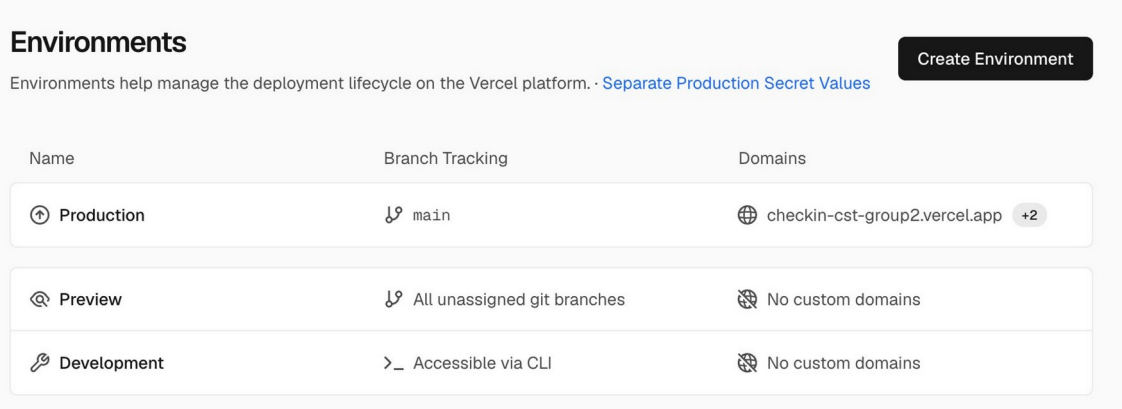
Ruleset — the three checks required before merging



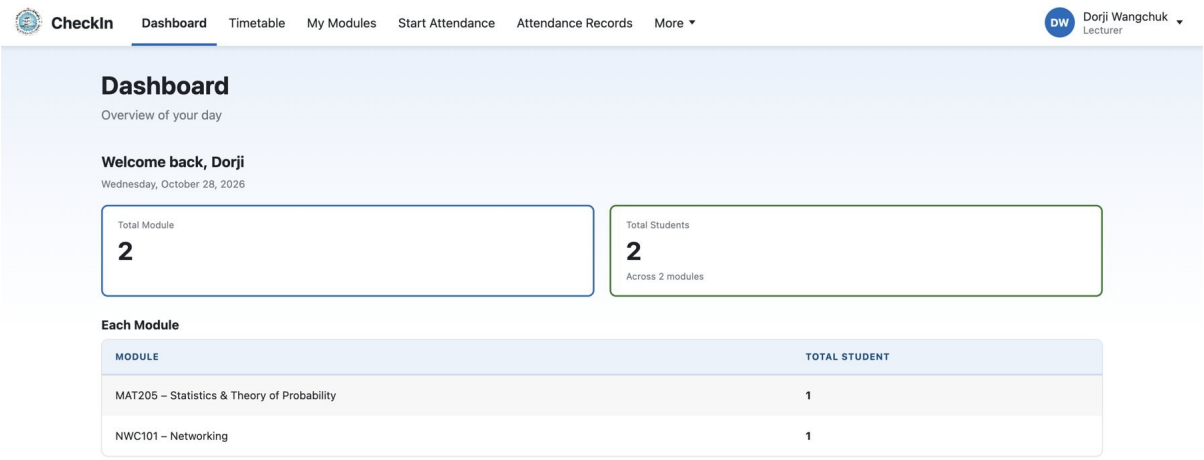
Render — checkin-backend live from main (commit 7b9d65c)



Vercel — frontend deployed to production



Vercel — production follows main, other branches get previews



The live site — tutor dashboard with real data from Supabase